

In the Name of God

Computer Programming Course Question Bank

Edition 1.4 (English Translation)

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This question set was written to meet the needs of students. An effort has been made to present questions ranging from a very basic level to an advanced level. As a result, the first questions in each section may seem very simple and repetitive — but they are necessary for beginner students — and the last questions in each section may be quite difficult and time-consuming, at the level of an advanced programming or data-structures course. Even so, they should be solvable using ordinary programming knowledge (not the knowledge of advanced courses or data structures). Please implement all of the questions so as to gain a complete, practical understanding of programming. To help make this collection more complete, please send new questions and your comments to: gholamali.irani@gmail.com, www.gClass.co

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Section One

Questions on if, for, and while

1. Write a program that takes a number from 1 to 365 and prints the current month.
2. Write a program that takes a number from 1 to 365 and prints the current day of the month.
3. Write a program that takes a number n representing an amount of money requested from an ATM, then breaks n down into denominations of 1, 5, 10, and 50 thousand Tomans.
4. Can you find and print the absolute value of a given number without using the if statement?
5. Can you find and print the min of two given numbers without using the if statement?
6. Can you find and print the max of two given numbers without using the if statement?
7. Write a program that prints the numbers 1 to 100.
8. Write a program that prints the sum of the numbers 1 to 100.
9. Write a program that prints the sum of the even numbers from 1 to 100.
10. Write a program that prints the sum of the odd numbers from 1 to 100.
11. Write a program that takes a number and prints all natural numbers less than or equal to it.
12. Write a program that takes a number and prints its divisors.
13. Write a program that takes a number and prints its odd divisors.
14. Write a program that takes a number and prints its even divisors.
15. Write a program that takes a number and prints the sum of its divisors.
16. Write a program that takes a number and prints the count of its divisors.
17. Write a program that takes a number and prints the sum of its odd divisors.
18. Write a program that takes a number and prints the count of its odd divisors.
19. Write a program that takes a number and prints the sum of its even divisors.
20. Write a program that takes a number and prints the count of its even divisors.
21. Write a program that takes a number and prints the average of its divisors.
22. Write a program that takes a number and prints the average of its even divisors.
23. Write a program that takes two numbers and computes and prints their GCD and LCM.
24. Write a program that takes a number and prints "Yes" if it is prime, "No" if it is not.
25. Write a program that takes a number and computes and prints its factorial.
26. Write a program that takes 10 numbers and computes and prints their average.
27. Write a program that takes 100 numbers and prints their min.
28. Write a program that takes 100 numbers and prints their max.

29. Write a program that takes 100 numbers and prints the second-smallest (min) among them.
30. Write a program that takes 100 numbers and prints the second-largest (max) among them.
31. Write a program that prints the prime numbers less than 1000.
32. Write a program that takes a number and prints its 2-digit divisors.
33. Write a program that takes a number and prints the sum of its 2-digit divisors.
34. Write a program that takes a number and prints the count of its 2-digit divisors.
35. Write a program that takes a number and prints the average of its 2-digit divisors.
36. Write a program that takes 100 numbers and prints how many of those 100 numbers are prime.
37. Write a program that takes 100 numbers and prints the sum of the prime numbers among those 100 numbers.
38. Write a program that takes 100 numbers and prints how many of those 100 numbers are odd.
39. Write a program that takes 100 numbers and prints the number that has the most divisors.
40. Write a program that prints the 50th prime number.
41. Write a program that prints the first 50 terms of the Fibonacci sequence.
42. Write a program that takes a number and prints the sum of its digits.
43. Write a program that takes a number and prints the sum of its even digits.
44. Write a program that takes a number and prints the sum of its odd digits.
45. Write a program that takes a number and prints the count of its digits.
46. Write a program that takes a number and prints the count of its even digits.
47. Write a program that takes a number and prints the count of its odd digits.
48. Write a program that takes a number and prints the product of its digits.
49. Write a program that takes a number and prints the product of its even digits.
50. Write a program that takes a number and prints the product of its odd digits.
51. Write a program that takes a number and prints the average of its digits.
52. Write a program that takes a number and prints the average of its even digits.
53. Write a program that takes a number and prints the average of its odd digits.
54. Write a program that takes a number and prints its reverse.
55. Write a program that takes a number and prints the reverse of its odd digits.
56. Write a program that takes a number and prints the reverse of its even digits.
57. Write a program that takes 2 numbers and prints their greatest common divisor.
58. Write a program that takes 2 numbers and prints their least common multiple.
59. Write a program that takes a number and decomposes it into prime factors.
60. Write a program that prints the 20th term of the Fibonacci series.

61. Write a program that prints the terms, among the first 50 terms of the Fibonacci series, that are prime.
62. Write a program that takes a number and determines whether it exists in the Fibonacci series.
63. Write a program that takes 100 numbers and prints how many of them exist in the Fibonacci series.
64. Write a program that takes 100 numbers and prints the sum of all digits of those 100 numbers.
65. Write a program that takes 100 numbers and prints the total digit count of those 100 numbers.
66. Write a program that takes 100 numbers and prints the average of the digit-averages of those numbers.
67. Write a program that takes 100 numbers and prints how many of them are equal to their own reverse.
68. Write a program that takes a number and removes its zeros.
69. Write a program that takes a number and prints the count of its prime divisors.
70. Write a program that takes a number and prints the sum of its prime divisors.
71. Write a program that takes a number and prints the average of its prime divisors.
72. Write a program that prints the first Fibonacci number that equals its own reverse.
73. Write a program that takes a number n and prints n rows of the pattern shown below (see the six sample shapes).
74. Write a program that prints a multiplication table.
75. Write a program that prints a base-8 multiplication table.
76. Write a program that first takes 100 numbers from the user, then prints how many of them are both prime and members of the Fibonacci series.
77. Write a program that takes a number and determines whether it is a perfect number.
78. Write a program that prints the perfect numbers less than 1000.
79. Write a program that takes 100 numbers and prints how many of them are perfect numbers.
80. Write a program that takes 100 numbers and prints how many of them are both perfect numbers and divisible by 2.
81. Write a program that takes a number and removes its zeros.
82. Write a program that takes a number and prints its equivalent in base 2.
83. Write a program that takes a number and prints its equivalent in base 8.
84. Write a program that takes two base-8 numbers, computes and prints their sum and product.
85. Write a program that takes a base-2 number and prints its equivalent in base 10.
86. Write a program that takes a base-8 number and prints its equivalent in base 10.
87. Write a program that takes a base-2 number and converts it to base 8.
88. Write a program that takes a number and prints its equivalent in base 16.
89. Write a program that takes a base-2 number and converts it directly to base 16.
90. Write a program that, for a given equation, takes x_1 and x_2 and computes and prints the value of the definite integral from x_1 to x_2 with precision 10^{-7} . (Methods: rectangular, trapezoidal, Simpson's, and

midpoint.)

91. Write a program that computes and prints, with precision 0.00001, the value of the integral $\int$ (from 1 to 5) of $\sin(x)/x \, dx$.
92. Write a program that, for a given nonlinear equation, takes two numbers x_1 and x_2 (where $f(x_1) \times f(x_2)$ is less than 0) and prints the root of the equation on that interval. (Methods: bisection, false position, secant, Newton, and fixed point.)
93. Write a program that, for a given nonlinear equation, computes and prints all of its roots.
94. Write a program that takes two numbers m and n , then computes and prints the value of $C(m, n)$ (the binomial coefficient).
95. Write a program that takes a number n and prints the n -th row of Pascal's triangle.
96. Write a program that takes a number n and prints n rows of Pascal's triangle.
97. Take a decimal number in base 10 and print its equivalent in base 2 (up to 5 fractional digits).
98. Take a decimal number in base 10 and print its equivalent in base 8 (up to 8 fractional digits).
99. Take a decimal number in base 2 and print its equivalent in base 10 (up to 5 fractional digits).
100. Take a decimal number in base 10 and print its equivalent in base 16 (up to 8 fractional digits).
101. Write a program that prints all 3-digit numbers that can be written as $xyz = x^3 + y^3 + z^3$ (for example, $153 = 1^3 + 5^3 + 3^3$).
102. Sum of the series: $1+2+3+\dots+100$
103. Sum of the series: $1-2+3-4+\dots-100$
104. Sum of the series: $1-3+5-7+\dots-99$
105. Sum of the series: $1/1 + 1/2 + 1/3 + \dots + 1/100$
106. Sum of the series: $1/1! + 1/2! + 1/3! + \dots + 1/100!$
107. Sum of the series: $1/20 + 2/19 + 3/18 + \dots + 20/1$
108. Sum of the series: $1/1! - 1/2! + 1/3! - \dots - 1/100!$
109. Sum of the series: $1/1! + 1/3! + 1/5! + \dots + 1/99!$
110. Sum of the series: $1 - 1/2! + 1/4! - \dots - 1/100!$
111. Sum of the series $1/1! + 1/2! + 1/3! + \dots$ with precision 10^{-7} .
112. Take x , then compute the series $1 + x/1! + x^2/2! + x^3/3! + \dots$ with precision 10^{-7} .
113. Take x , then compute the series $1 + x^2/2! + x^4/4! + \dots$ with precision 10^{-7} .
114. Take x , then compute the series $x - x^3/3! + x^5/5! - \dots$ with precision 10^{-7} .
115. Sum of the series: $1!/1^1 + 2!/2^2 + 3!/3^3 + \dots + 100!/100^{100}$
116. Sum of the series $1!/1^1 + 2!/2^2 + 3!/3^3 + 4!/4^4 + \dots$ with precision 10^{-7} .
117. Sum of the series $1!/1^1 - 3!/3^3 + 5!/5^5 - 7!/7^7 + \dots$ with precision 10^{-10} .
118. Sum of the continued-fraction series below, with precision 10^{-10} : $1 + 1/(2 + 1/(2 + 1/(2 + \dots)))$

119. Sum of the series below, with precision 10^{-10} : $\ln(x) = 2[(x-1)/(x+1) + (1/3)((x-1)/(x+1))^3 + (1/5)((x-1)/(x+1))^5 + \dots]$
120. Look up the series related to $\sin(x)$, $\cos(x)$, e^x , $\ln(x)$, $\sqrt{x}$, etc. that arise from the Taylor expansion, then write a program that takes x and computes the desired series with precision 10^{-7} and prints it. (You may use http://en.wikipedia.org/wiki/Taylor_series for the series formulas.)

Good luck and be happy – Irani

Section Two

Questions on One-Dimensional Arrays

121. Write a program that takes 5 numbers and prints them from last to first.
122. Write a program that takes 20 numbers and prints them from last to first.
123. Write a program that takes 20 numbers and prints the numbers that are greater than the average of the 20 entered numbers.
124. Write a program that takes 20 numbers and prints how many of the numbers are greater than the average.
125. Write a program that takes 20 numbers and prints the numbers that are prime and greater than the average.
126. Write a program that takes 20 numbers and prints the numbers that belong to the Fibonacci series and are greater than the average.
127. Write a program that takes n numbers and prints them from last to first.
128. Write a program that takes n numbers, then takes a number m, and prints how many times m occurs among the n numbers.
129. Write a program that takes n numbers, then takes a number m, and prints the positions of m among the n numbers.
130. Write a program that takes a long number and prints its equivalent in base 2.
131. Write a program that takes 20 numbers and removes the 5th element (shifting the numbers and reducing the total count), then prints the remaining numbers.
132. Write a program that takes n numbers, then takes a number m, and removes all occurrences of m among the n numbers, then prints the remaining numbers.
133. Write a program that takes n numbers, then takes two numbers p1 and p2, and removes the elements from position p1 through p2, then prints the remaining numbers.
134. Write a program that takes 10 numbers, then shifts them 3 positions forward (circularly), then prints the remaining numbers.
135. Write a program that takes n numbers, then takes a number m, and shifts the n numbers m positions forward (circularly), then prints the remaining numbers.
136. Write a program that takes n numbers, stores them in an array, then rearranges the array's elements based on the middle (first with last, second with second-to-last, etc.), and finally prints the resulting array.
137. Write a program that takes n numbers and stores them in an array. Then it stores the positive and negative numbers in separate arrays and finally prints each of the resulting arrays.
138. Write a program that takes n numbers and stores them in an array. Then it stores the prime and non-prime numbers in separate arrays and finally prints each of the resulting arrays.
139. Write a program that takes n numbers and stores them in an array. Then it stores the prime, composite, and Fibonacci numbers in separate arrays and finally prints each of the resulting arrays.

140. Write a program that takes n numbers, removes the even numbers among them, and prints the final numbers.
141. Write a program that takes n numbers, removes the prime numbers among them, and prints the final numbers.
142. Write a program that takes n numbers, then takes two numbers m and k, and inserts the number m at position k of the array.
143. Write a program that takes n numbers, then places the divisor count of each number right after it, and finally prints all the numbers in the array.
144. Write a program that takes n numbers, then places the digit count of each number right after it, and finally prints all the numbers in the array.
145. Write a program that takes n numbers and m other numbers, then finds and prints the numbers common to both lists.
146. Write a program that takes n numbers, then takes a number m, and prints how many times m occurs among the n numbers.
147. Write a program that takes n numbers and places the repeat count of each number into a corresponding second array.
148. Write a program that takes n numbers and prints the number that occurs most frequently.
149. Write a program that takes n numbers and prints the index of the element whose value occurs most frequently.
150. Write a program that takes n numbers, removes its duplicate values, and prints the result.
151. Write a program that takes n numbers and prints the numbers that occur most frequently (there may be more than one).
152. Write a program that prints a random number.
153. Write a program that prints a random integer between 1 and 100.
154. Write a program that prints a random integer between a and b.
155. Write a program that generates 20 random numbers between 1 and 100, places them in an array, and prints the array.
156. Write a program that generates and prints 20 non-repeating random numbers between 1 and 100.
157. Write a program that takes n numbers between 1 and 10, then prints how many times each of them occurs.
158. Using one-dimensional arrays, write a program that takes a number n and prints the n-th row of Pascal's triangle.
159. Using one-dimensional arrays, write a program that takes a number n and prints n rows of Pascal's triangle.
160. Write a program that takes n numbers and determines whether they are sorted.
161. Write a program that takes n numbers and prints their median.
162. Write a program that takes n numbers and sorts them. (Bubble sort, insertion sort, selection sort.)

163. Write a program that takes two groups of numbers, of size m and n , and places their sorted merge into a third array, then prints the resulting array. (Merging two sorted arrays.)
164. Write a program that takes n sorted numbers and a number m , then searches for m among the n numbers using binary search.
165. Write a program that takes n sorted numbers, then takes a number m , and searches for it among the n numbers using ternary search.
166. Write a program that computes and prints $1000!$.
167. Write a program that computes and prints $100000!$.
168. Write a program that computes and prints $1000000!$.
169. Write a program that computes and prints 2^{100} .
170. Write a program that takes two long numbers m and n , then computes and prints m^n .
171. Write a program that computes and prints $(1000!)^3$.
172. Write a program that takes two very large integers (e.g. 100 digits) and prints their sum.
173. Write a program that takes two very large integers (e.g. 100 digits) and prints their difference.
174. Write a program that takes two very large integers (e.g. 100 digits) and prints their product.
175. Write a program that takes two very large integers (e.g. 100 digits) and prints their integer quotient.
176. Write a program that takes two very large integers (e.g. 100 digits) and prints their integer remainder.
177. Write a program that takes two very large integers (e.g. 100 digits) and computes and prints the result of their decimal (fractional) division.
178. Compute the series: $1^1 + 2^2 + 3^3 + \dots + 100^{100}$
179. Compute the series: $1! + 2! + 3! + \dots + 100!$
180. Write a program that computes and prints the value of $(2^{64})!$.

Good luck and be happy – Irani

Section Three

Questions on Strings

181. Write a program that takes a letter and displays it in uppercase (print A instead of a).
182. Write a program that takes a letter and displays it in lowercase (print a instead of A).
183. Write a program that takes a character and prints its ASCII code.
184. Write a program that prints all ASCII codes from 0 to 255 together with their equivalent characters, formatted as a table.
185. Write a program that takes a number and prints its equivalent in base 16.
186. Write a program that takes a string and prints it.
187. Write a program that takes a base-16 number and prints its equivalent in base 10.
188. Write a program that takes a string, converts all of its uppercase letters to lowercase and all of its lowercase letters to uppercase, and prints it.
189. Write a program that takes a string and prints the average of the digits found in it.
190. Write a program that takes a string, removes all the digits found in it, and prints the result.
191. Write a program that takes a string S and a character c, and prints the positions where c occurs in S.
192. Write a program that takes two strings S1 and S2, finds the number of occurrences of S2 within S1, and prints it.
193. Write a program that takes a string, capitalizes the first letter of every word in it, and prints the result.
194. strtoint: Write a program that reads a number of up to 10 digits as a string from input, then derives its numeric value from the string and finally prints it. ("538" becomes 538)
195. charAt: Write a program that takes a string and a number n, and prints the character at position n.
196. Reverse: Write a program that takes a string and prints it from last character to first.
197. ToLower: Write a program that takes a string, converts all of its letters to lowercase, and prints it.
198. ToUpper: Write a program that takes a string, converts all of its letters to uppercase, and prints it.
199. SetW: Write a program that takes a string, then a number n, and inserts space characters at the beginning of the string until its length reaches n, then prints the result.
200. SubString: Write a program that takes a string S and two numbers n1 and n2, copies the portion of S from position n1 to position n2 as a new string, and prints the result.
201. Ltrim: Write a program that takes a string and removes the spaces on its left side.
202. Rtrim: Write a program that takes a string and removes the spaces on its right side.
203. Trim: Write a program that takes a string and removes the spaces at its beginning and end.
204. FullTrim: Write a program that takes a string, removes the spaces at its beginning and end, then reduces every run of more than one internal space down to a single space, and prints the result.

205. IndexOf: Write a program that takes a string S, then a character c and a number n, and finds and prints the first occurrence of c in S starting from position n (print -1 if not found).
206. IndexOf2: Write a program that takes a string S1, then another string S2 and a number n, and finds and prints the first occurrence of S2 within S1 starting from position n (print -1 if not found).
207. Compare: Write a program that takes two strings, compares them for equality, and prints the result.
208. Compare2: Write a program that takes two strings and compares them for equality while ignoring letter case, then prints the result.
209. Remove: Write a program that takes a string and a character, and removes every occurrence of that character from the string.
210. Remove2: Write a program that takes two strings S1 and S2, and removes every occurrence of S2 within S1.
211. Replace: Write a program that takes three strings S1, S2, and S3, and replaces every occurrence of S2 within S1 with S3.
212. Split: Write a program that takes a string S and a character c, splits S into separate substrings wherever c occurs, and prints the resulting substrings.
213. Write a program that takes a string and prints all of its subsets.
214. Write a program that takes a string of length n and prints its m-element subsets (where m is less than n).
215. Write a program that takes two strings s1 and s2 and prints the number of occurrences of s2 within s1.
216. Write a program that takes a string and prints its repeated (duplicate) characters.
217. Write a program that takes a string and prints its repeated words.
218. Write a program that takes a string and prints its complement string (relative to the full reference set of English letters).
219. Write a program that takes two strings s1, s2 and prints their Cartesian product.
220. Write a program that takes two strings s1, s2 and prints the characters common to both.
221. Write a program that takes two strings s1, s2 and prints the intersection of their words.
222. Write a program that takes a string and prints all of its permutations.
223. Write a program that takes 3 strings and prints the (combined) permutations of the given strings.
224. First define a two-way (reversible) encryption algorithm, then using it write a program that takes a string and encodes it, or takes an encoded string and decodes it.
225. Write a program that takes a positive integer (at most 9 digits) from the user and prints its Persian word form.
226. Write a program that takes a positive integer (at most 9 digits) from the user and prints its English word form.
227. Write a program that takes a decimal number (at most 9 integer digits and 8 fractional digits) from the user and prints its Persian word form.
228. Write a program that takes two very large integers as strings from input and computes and prints their sum.

229. Write a program that takes two very large integers as strings from input, then computes their difference and prints it.
230. Write a program that takes two very large integers as strings from input, then computes their product and prints it.
231. Write a program that takes two very large integers as strings, then computes their integer division and prints it.
232. Write a program that takes two very large integers as strings, then computes their integer remainder and prints it.
233. Compute the series: $1^1 + 2^2 + 3^3 + \dots + 100^{100}$
234. Compute the series: $1! + 2! + 3! + \dots + 100!$
235. Write a program that computes and prints the value of $(2^{64})!$.
236. Write a program that takes two polynomials as strings and prints their sum.
237. Write a program that takes two polynomials as strings and prints their difference.
238. Write a program that takes two polynomials as strings and prints their product.
239. Write a program that takes two polynomials as strings, computes their quotient and remainder, and prints them.
240. Write a program that takes a polynomial as a string and prints its derivative.
241. Write a program that takes a polynomial as a string and prints its integral.
242. Write a program that reads a mathematical expression containing digits and the "+" sign, such as "7+534+12", as a string, then computes and prints its numeric value.
243. Redo the previous problem, also adding the "-" sign.
244. Write a program that reads a mathematical expression containing digits and the "+" and "×" signs, such as "173+34×51+12", as a string, then computes and prints its numeric value.
245. Write a program that reads a mathematical expression containing digits, the "+" and "×" signs, and the "(" and ")" symbols, as a string, then computes and prints its numeric value.
246. Redo the expression-evaluation problems above for the case of decimal numbers and negative numbers as well.
247. Write a program that takes a number and prints its equivalent in base 32.
248. Write a program that takes a sentence, capitalizes the first letter of every word, and prints the result.
249. Write a program that takes a sentence and removes its repeated words.
250. Write a program that reads a mathematical equation and plots its graph.

Section Four

Questions on Two-Dimensional Arrays

251. Write a program that takes a 10×10 matrix from the user.
252. Write a program that takes a 10×10 matrix and prints it.
253. Write a program that takes m and n from the user, generates an identity matrix, and prints it.
254. Write a program that takes a 10×10 matrix and prints only its main diagonal.
255. Write a program that takes a 10×10 matrix and prints only its secondary (anti-) diagonal.
256. Write a program that takes a 10×10 matrix and prints the average of the numbers on its main diagonal.
257. Write a program that takes a 10×10 matrix and prints only its even-numbered rows.
258. Write a program that takes a 10×10 matrix and prints the max of all the numbers in it.
259. Write a program that takes a 10×10 matrix and prints the max of each row.
260. Write a program that takes a 10×10 matrix and prints the min of each column.
261. Write a program that takes a 10×10 matrix and two numbers m and n , then swaps rows m and n and prints the resulting matrix.
262. Write a program that takes a 10×10 matrix and two numbers m and n , then swaps columns m and n and prints the resulting matrix.
263. Write a program that takes two numbers m and n , then takes an $m \times n$ matrix, places its transpose into another matrix, and prints it.
264. Write a program that takes two numbers m and n , takes an $m \times n$ matrix, zeroes out the lower triangle of the matrix, and prints the result.
265. Write a program that takes two numbers m and n , takes an $m \times n$ matrix, zeroes out the upper triangle of the matrix, and prints the result.
266. Write a program that takes two numbers m and n , then an $m \times n$ matrix, then another number P , and prints the coordinates at which P occurs in the matrix.
267. Write a program that takes 100 numbers and places them into a 100-element array, then copies the 100 numbers, in order, row by row, into a 10×10 matrix.
268. Write a program that takes 45 numbers and places them into a 45-element array, then copies the 45 numbers, in order, row by row, into a 9×5 matrix.
269. Write a program that takes 45 numbers and places them into a 45-element array, then copies the 45 numbers, in order, row by row, into a 5×9 matrix.
270. Write a program that takes 45 numbers and places them into a 45-element array, then copies the 45 numbers, in order, column by column, into a 9×5 matrix.
271. Write a program that takes 45 numbers and places them into a 45-element array, then copies the 45 numbers, in order, column by column, into a 5×9 matrix.

272. Write a program that takes two numbers m and n , then an $m \times n$ matrix, sorts each row of the matrix, and prints the result.
273. Write a program that takes two numbers m and n , then an $m \times n$ matrix, sorts each column of the matrix, and prints the result.
274. Write a program that takes two numbers m and n , then two $m \times n$ matrices, computes their sum, places it in a third matrix, and prints the result.
275. Write a program that takes two numbers m and n , then an $m \times n$ matrix, then another number P , adds P to every cell of the matrix, and prints the final matrix.
276. Write a program that takes two numbers m and n , then an $m \times n$ matrix, then another number P , multiplies every cell of the matrix by P , and prints the final matrix.
277. Write a program that takes three numbers m , n , and k , then two matrices of size $m \times k$ and $k \times n$, computes their product, and prints the resulting $m \times n$ matrix.
278. Write a program that takes a number n , then an $n \times n$ matrix, and computes and prints its inverse.
279. Write a program that takes a number n , then an $n \times (n+1)$ matrix as the coefficients of an $n \times n$ linear system, computes its n roots, and prints them. (Methods: Gauss, Gauss-Jordan, LU decomposition.)
280. Write a program that takes a matrix and computes and prints its eigenvalues.

Good luck and be happy – Irani

Section Five

Questions on Methods

281. Write a method that returns the min of two int numbers – `public static int max(int x, int y);`
282. Write a method that returns the min of two float numbers – `public static float max(float x, float y);`
283. Write a method that returns the absolute value of an int number – `public static int abs(int x);`
284. Write a method that returns the absolute value of a float number – `public static float abs(float x);`
285. Most of the programs in Section 1 (the if / for / while section) can also be implemented using methods. For example, consider the following program: "Write a program that takes a number and prints the count of its divisors." This can be restated as: "Write a method that takes a number as the method's input, computes the count of its divisors, and returns it as the method's output."
286. Most of the programs in Section 2 (the one-dimensional arrays section) can also be implemented using methods. For example, the program that merges two sorted arrays can be restated as: "Write a method that receives two sorted arrays as input, places their sorted merge into a third array, and returns it as the method's output."
287. Likewise, most of the questions in Sections 1, 2, 3, and 4 can be phrased as methods. Below are several examples.
288. Write a method that takes a string as input, converts its lowercase letters to uppercase and its uppercase letters to lowercase, and returns the result as the method's output.
289. Write a method that takes two strings S1 and S2 as input, finds and removes every occurrence of S2 within S1, then returns the result as the method's output.
290. Write a method that takes a string as input, capitalizes the first letter of every word, and returns the result as the method's output.
291. Write a method that takes a string as input, splits it into words, and returns them as an array of strings.
292. Write a method that takes a number n ($1 \leq n \leq 1000$) and an array as input, computes $n!$, stores it in the array, and returns the length of the resulting array as the method's output.
293. Write a method that takes three arrays, together with the lengths of the first two, as input, computes the product of the first two arrays, places the result in the third array, and returns the length of the resulting array as the method's output.
294. Write a method that takes a string (containing an ordinary decimal number such as "-328.098" or "101.009") as input, converts the decimal number found in the string into a floating-point number, and returns it as the method's output.
295. Method BigSum, which takes two large positive numbers as strings and returns their sum as a string.
296. Method BigMinus, which takes two large positive numbers as strings, where the first is larger, and returns their difference as a string.
297. Method BigMultiple, which takes two large positive numbers as strings and returns their product as a string.

298. Method BigDiv, which takes two large numbers as strings and returns the result of their integer division as a string.
299. Method BigMod, which takes two large numbers as strings and returns the remainder of their integer division as a string.
300. Method BigCompare, which takes two large positive numbers as strings and returns the result of comparing them as 1, -1, or 0.
301. Method PSum, which takes two polynomials as strings and returns their sum as a string.
302. Method PMinus, which takes two polynomials as strings and returns their difference as a string.
303. Method PMultiple, which takes two polynomials as strings and returns their product as a string.
304. Method PDiv, which takes two polynomials as strings and returns their quotient as a string.
305. Method PMod, which takes two polynomials as strings and returns their remainder as a string.
306. Write a method named MLPlusMinus that takes a mathematical expression as a string – containing numbers (positive or negative, of type long) and the "-" and "+" signs – and returns its result as a long number. (e.g. "-438-2+23" or "329+9-9128")
307. Write a method that takes a mathematical expression as a string containing numbers (positive or negative, long) and the "-", "+", "×", and "/" signs, and returns its result as a double.
308. Write a method that takes a mathematical expression as a string containing numbers (positive or negative, long) and the "-", "+", "×", "/", and "^" signs, and returns its result as a double.
309. Write a method that takes a mathematical expression as a string containing numbers (positive or negative, long) and the "-", "+", "×", "/", "^", "(", and ")" signs, and returns its result as a double.
310. Write a method that takes a mathematical expression as a string containing decimal numbers (positive or negative, double), the "-", "+", "×", "/", "^", "(", and ")" signs, and functions such as Sin, Cos, Tan, Log, Ln, Sqr, and Exp, and returns its result as a double.
311. Implement all of the following String-related methods: StrToInt, CharAt, Reverse, ToLower, ToUpper, SetW, SubString, LTrim, RTrim, Trim, FullTrim, IndexOf, Compare, Remove, Replace, Split.

Good luck and be happy – Irani

Section Six

Sample Final Project Lists

The original document lists sample end-of-term programming projects assigned in several different academic terms at the University of Bonab. Each table below corresponds to one term.

Programming Course Projects – Computer Engineering Group (First Semester 1389)

Project Name	Description	Score
Numerical Integration Project	The project must be a single program that computes and prints a definite integral in a specific interval using all 7 of the methods below. Use of functions for each method is mandatory. Methods: rectangular, trapezoidal, Simpson's, midpoint, Romberg, 2-point Gauss, and 3-point Gauss quadrature.	1
Polynomial Root-Finding Project	Compute and print the roots of a polynomial.	1
Linear Systems Project	Take a matrix of size $n \times (n+1)$ and solve the resulting system using the 4 methods below. You may write a separate program for each method. Methods: Gauss, Gauss-Jordan, Gauss with pivoting, and LU decomposition.	2
Matrix Inversion Project	Take an $n \times n$ matrix, compute and print its inverse.	1
Big-Number Calculator Project	1) Take two large numbers and an operator among *, -, + and print the result. 2) Take two large numbers and print the result of their division. 3) For an ordinary number, compute and print its factorial. 4) Take two ordinary numbers and print the result of raising one to the power of the other. The whole project must be a single program, and the use of functions is mandatory for each operation.	2
Compute and Store Factorials of Large Numbers	Write a program that computes 100000!, 200000!, 300000!, ..., 1000000! and stores each in a separate file. Then it takes a number and, by reading the nearest stored value from file, computes the factorial of that number.	2

List of Programming Course Projects – Computer Engineering Group, University of Bonab (First Semester 1390)

Group	Description	Item	Score
Numerical Analysis Project (NAP)	Numerical integration with the rectangular, trapezoidal, Simpson's, and midpoint methods: for a specific equation predefined in the program (with other parameters such as start point, end point, precision, etc.), compute the definite integral over the given interval and print it. The program should be menu-driven, and one of the menu options should print the results of all methods together. Solving nonlinear equations with false-position, fixed-point, Newton, and secant methods: same as above. Solving linear systems with Gauss and Gauss-Jordan methods: the program should read an $n \times (n+1)$ matrix as a linear system from input and solve it with the Gauss or Gauss-Jordan method, printing the results. Matrix operations, including addition and multiplication of two matrices and the inverse of a matrix: menu-driven; take two matrices for addition/multiplication and perform the requested operation, and for inversion take a matrix and print its inverse.	P11 / P12 / P13 / P14	1 / 1 / 4 / 4 (total 10)
Big Numbers Calculator (BNC)	Addition, subtraction, and multiplication of two large numbers: take two large numbers from the user and perform the requested operation. Division: take two large numbers, perform integer division, and print the quotient and remainder. You must use the addition, subtraction, multiplication, and comparison functions, etc. in this program. Power and factorial of large numbers: similar to the first item.	P21 / P22 / P23	4 / 15 / 1 (total 20)
Equation Lab (EL)	Addition, subtraction, and multiplication of polynomials: take two polynomials from input and print their sum, difference, and product. Division of polynomials: take two polynomials, divide them, and print the quotient and remainder. Derivative and integral of polynomials: take a polynomial and compute and print its derivative and integral.	P31 / P32 / P33	3 / 5 / 2 (total 10)
Math Lab (ML)	Evaluating simple math expressions: a string expression containing numbers and + / -; compute and print its value. Evaluating ordinary math expressions: a string expression containing numbers and + / - / $\times$ / $\div$; compute and print its value. Evaluating parenthesized math expressions: a string expression containing numbers, + / - / $\times$ / $\div$, and (); compute and print its value. Evaluating math functions: a string function containing numbers (which may be multi-digit), + / - / $\times$ / $\div$ / ^, (), and functions such as sin, cos, tan, log (i.e. Ln), sqr (square root), and exp – e.g. $\text{Log}(x \cdot \sin(2 \cdot x)) - \exp(x^2 - 1) + 2 \cdot x^{12} - 10 \cdot x + \text{sqr}(x) \cdot x^3$ – then take a number such as 2.5, substitute it into the function, compute, and print the result.	P41 / P42 / P43 / P44	5 / 5 / 5 / 15 (total 30)

Group	Description	Item	Score
University System (US)	Student and course record system: the program must support: 1) Adding student records (student ID, first name, last name, ID number, address), stored in a file sorted by student ID. 2) Editing, deleting, and simple/advanced searching of all student records. 3) Adding course records (course code, course name, theory units, practical units). 4) Editing, deleting, and simple/advanced searching of all course records. 5) Recording, editing, and searching each student's course registration.	P51	30 (total 30)

List of Programming Course Projects – Computer Engineering Group (First Semester 1391) – University of Bonab

Group	Description	Item	Score
Numerical Analysis Project (NAP)	Solving linear systems with Gauss and Gauss-Jordan methods: the program should read an $n \times (n+1)$ matrix as a linear system and solve it with the Gauss and Gauss-Jordan methods, printing the results.	P11	10 (total 10)
Big Numbers Calculator (BNC)	Addition, subtraction, and multiplication of two large numbers. Division: integer division with quotient and remainder, using the addition/subtraction/multiplication/comparison functions etc. Power and factorial of large numbers , similar to the first item.	P21 / P22 / P23	-3 / -6 / -1 (total -10)
Polynomial Lab (PL)	Addition, subtraction, and multiplication of two polynomials. Division of polynomials (quotient and remainder). Derivative and integral of a polynomial.	P31 / P32 / P33	-3 / -5 / -2 (total -10)
Math Lab (ML)	Evaluating parenthesized math expressions. Evaluating math functions (as a string of digits, + / - / $\times$ / $\div$ / ^, (), and functions sin, cos, tan, log/Ln, sqr, exp) – sample: $\text{Log}(x \cdot \sin(2 \cdot x)) - \exp(x^2 - 1) + 2 \cdot x^{12} - 10 \cdot x + \text{sqr}(x) \cdot x^3$ – then substitute a value such as 2.5 and print the result. Find all zeros of a function taken as in P42.	P41 / P42 / P43	5 / 25 / 10 (total 40)
Library System (LS)	University library record-keeping system: 1) Add, edit, delete, and simple/advanced search of student records (student ID, first name, last name, national ID, phone number, address), stored in a file sorted by student ID. (10 pts) 2) Add, edit, delete, and simple/advanced search of book records (book number, title, author, translator, print run), stored in a file sorted by book number. (10 pts) 3) Recording and searching book loans and returns in the system. (15 pts)	P51	35 (total 35)
Very Big (VB)	Write a program that computes $n!$ (for n up to about 60,000) and stores it in a file.	P61	+20 (bonus)
Quick Fact (QF)	Write a program that computes $100000!$, $200000!$, ..., $1000000!$ and stores each in a separate file. (5 pts) Take a number, find the nearest value from the stored files, compute its factorial, and store the result in a file named after the given number. (10 pts)	P71 / P72	5 / 10 (total 15)

List of Fundamentals-of-Computer-and-Programming Course Projects – Computer Engineering Group (First Semester 1392) – University of Bonab

Group	Item	Description	Score
P1: Big Numbers (BN)	P11	First implement the following methods: • BigSum – takes two large positive numbers as strings, returns their sum as a string. • BigMinus – takes two large positive numbers as strings (first $\geq$ second), returns their difference as a string. • BigMultiple – takes two large positive numbers as strings, returns their product as a string. • BigCompare – takes two large positive numbers as strings, returns 1, -1, or 0 as their comparison result. Based on the methods above and any additional methods you define, implement the following and demonstrate their use with an example: BigDiv – returns the integer quotient of two large numbers given as strings. (5 pts) BigMod – returns the integer remainder of two large numbers given as strings. (-5 pts)	5 (total 10)
	P12	Implement a calculator for large numbers (based on the P11 methods and any others you define). The user enters a number, then an operator, then a second number, and the result is printed. The operator may be $- + \times / \% ^ !$ (for ! no second number is needed).	5
P2: Polynomial Lab (PL)	P21	PSum, PMinus, PMultiple methods – take two polynomials as strings, return their sum / difference / product as a string.	5 (total 10)
	P22	Based on P21 and any additional methods: PDiv – returns the quotient of two polynomial strings. (5 pts) PMod – returns the remainder of two polynomial strings. (-5 pts)	5
P3: Math Lab (ML)	P31	Method ML31: takes a math expression as a string containing whole numbers and the operators - and +, returns the result as a long number. (e.g. "-438-2+23" or "329+9-9128")	-5
	P32	Method ML32: as ML31 but numbers can also be decimal.	5
	P33	Method ML33: as ML32 but with (and) added.	5
	P34	Method ML34: as ML33 but with $\times$, /, and $^$ added.	10
	P35	Method ML35: as ML34 but able to handle functions such as Sin, Cos, Tan, Log, Ln, Sqr, Exp (no functions implemented: -3 pts each).	10 (group total 30)
P4: (ML2)	P41	Based on method P35, write a program that takes a nonlinear equation as a string and prints all of its roots.	10 (total 10)
P5: (Files)	P51	Restaurant information system, with the following capabilities: 1) Add, edit, delete, and simple/advanced search of member records (membership number, first name, last name, national ID, phone number, address), stored in a file sorted by membership number. (5 pts) 2) Add, edit, delete, and simple/advanced search of food and dessert items (item number, title, price, quantity on hand, minimum stock threshold), stored in a file sorted by item number. (5 pts) 3) Recording sales invoices, searching recorded invoices, and deactivating invoices in the system. (10 pts) 4) Displaying statistics (list of available food items, best-selling items, members with the highest purchases, today's cash-register total, items below their minimum stock threshold, etc.). (10 pts)	30 (total 30)
P6: (Quick Fact)	P61	Write a program that computes 100000!, 200000!, ..., 1000000! and stores each in a separate file.	5

Group	Item	Description	Score
	P62	Take a number, find the nearest stored value from file, compute its factorial, and store the result in a file named after the given number.	5 (total 10)

List of Fundamentals-of-Computer-and-Programming Course Projects – Computer Engineering Group (First Semester 1393) – University of Bonab

This term's project list (Big Numbers, Polynomial Lab, Math Lab, and Files groups) follows the same structure as the 1392 list above, with the following notable difference:

Group	Item	Description	Score
P4: Big Bases (BB)	P41	Write a program that takes a large number as a string and prints its equivalent in base 16 and base 32.	5
	P42	Using a suitable coding table, extend P41 to base 100.	5
	P43	Extend P42 to all bases less than 1,000,000.	5
	P44	Write addition and multiplication of two large numbers in large bases (up to 1,000,000).	15 (total 30)
P5: (Files)	P51	Bank record-keeping system, with the following capabilities: 1) Add, edit, delete, and search all customer records (account number, balance, first name, last name, national ID, phone number, address, account-opening date), stored in a file sorted by account number. (5 pts) 2) Record deposit information (deposit slip number, account number, date, depositor's national ID/name, amount), stored in a separate file. (5 pts) 3) Record withdrawal information (withdrawal slip number, account number, date, amount), stored in a separate file. (5 pts) 4) Block/unblock a given account (disable withdrawals); delete a withdrawal or deposit record (isDelete). (5 pts) 5) Display statistical reports of a specific account's transactions over a given date range, sorted by date. (5 pts) 6) Display statistical reports of the whole bank's transactions (sum of all accounts) over a given date range. (5 pts)	30 (total 30)

List of Fundamentals-of-Computer-and-Programming Course Projects – Computer Engineering Group (First Semester 1394) – University of Bonab

Group	Item	Description	Score
P1: Big Numbers (BN)	P11	BigSum, BigMinus, BigMultiple, BigCompare methods (as before). (10 pts) BigDiv (5 pts) and BigMod (-5 pts), based on the methods above. Write a simple test program based on the methods above.	10 (total 35)
	P12	Write a method that takes a large positive number as a string and returns its equivalent in base 16.	5
	P13	Write a program that computes the series $1^1+2^2+3^3+\dots+100^{100}$ in base 16 and prints it. (-5 pts)	-
	P13'	Write a program that computes $(2^{64})!$ and stores the result in a text file.	10
	P14	Write a program that computes the first million-digit prime number and stores the result in a text file.	10
P2: Polynomial Lab (PL)	P21	PSum, PMinus, PMultiple methods for polynomial strings.	5 (total 10)
	P22	Based on P21 and any additional methods: PDiv (5 pts), PMod (-5 pts).	5
P3: Math Lab (ML)	P31	Method ML31: takes a math expression as a string containing ordinary decimal numbers and the operators - and +, returns a decimal result. (e.g. "-438.001078-0.2021+2.3101003")	5 (total 25)
	P32	Method ML32: using ML31, add the operators $\times$, /, and $^$ (respecting operator precedence).	5
	P33	Method ML33: using ML31/ML32, add () (respecting operator precedence).	5
	P34	A method named ML: using the methods above, plus functions such as Sin, Cos, Tan, Log, Ln, Sqr, Exp, evaluate a complete math expression and return a decimal result.	10
P4: Numerical Analysis Project (NAP)	P41	Read an $n \times n$ matrix of decimal numbers from a text file (m.txt, first line gives n), compute its inverse, and store it in m-Inverse.txt.	5 (total 10)
	P42	Read an $n \times (n+1)$ matrix of decimal numbers (linear system coefficients) from m1.txt (first line gives n), solve the system by the Gauss or Gauss-Jordan method, and store the results in m1-Gauss.txt and m1-Jordan.txt.	5
P5: (Files)	P51	Room/dormitory expense-accounting system, with the following capabilities: 1) Member records: add, edit, delete, and search member data (unique member/student ID, first name, last name, phone number, registration date), stored in a file sorted by member ID. (5 pts) 2) Expense records: record a unique expense-voucher number, a full description of items and reasons, the paying member's ID, payment date, total amount, and which members share the expense, stored in a separate file. (10 pts) 3) Display the total of all expenses, sorted and tabulated. (5 pts)	20 (total 20)

Section Seven

Sample Final-Term Exam Questions

The following tables reproduce sample final-exam question sets from several terms. Where the original instructs students to “explain each part of the program in Persian,” that instruction has been kept as “explain each part of the program in your own words,” since these were originally Persian-language exams.

First Semester 1391

#	Question	Score
1	Write a program that takes 100 numbers and prints the count of numbers that are both part of the Fibonacci series and greater than the average of the 100 numbers entered. Explain each part of the written program in your own words. (5 pts for not doing so)	10
2	Write a program that computes and prints the following series with precision 0.0000001. (The program must not raise an Error.) $1 - 2^2/3! + 3^3/5! - 4^4/7! + 5^5/9! - \dots - 100^{100}/199!$ Explain each part of the written program in your own words. (5 pts for not doing so)	10
3	Write a method that takes two strings S1 and S2 as input, finds and removes every occurrence of S2 within S1, then returns the result as the method's output. Explain each part of the written program in your own words. (5 pts for not doing so)	10
4	Write a program that reads a 10×10 matrix from a file named A.txt, sorts each of the matrix's columns, and stores the result in file B.txt. Explain each part of the written program in your own words. (5 pts for not doing so)	10
5	Write a program that takes 100 numbers, stores them sorted in an array, then takes a number M and searches for it in the array. The search method must resemble binary search, except that at each step the whole array is split into 3 parts, and after comparison 2 of the 3 parts are discarded, repeating for the remaining part (ternary search). Explain each part of the program and the variables used, in your own words. (5 pts for not doing so)	10
6	Write a program that takes a math expression, containing +, ×, (), and positive numbers, as a string from input, then prints its value. You may define different methods for different tasks and use them. (+10 pts) Explain each part of the program and the variables used, in your own words. (10 pts for not doing so)	25
7	Assume the following methods already exist: BigDiv, BigMod, BigSum, BigMinus, BigMultiple, BigCompare (as described in earlier sections). Write a program that takes a large number and decomposes it into prime factors. You may define other methods as needed. Explain each part of the program and the variables used, in your own words. (10 pts for not doing so) <i>Important note: the numbers and exponents obtained from the prime factorization may themselves be large numbers.</i>	25

Second Semester 1391

#	Question	Score
1	Write a program that takes a number, then prints the count of divisors that are greater than the average of that number's divisors.	10
2	Write a program that computes and prints the following series with precision 0.0000001. (The program must not raise an Error.) $1/1! - 3^3/3! + 5^5/5! - 7^7/7! + \dots$	10
3	Write a program that computes and prints <u>all</u> roots of the nonlinear equation $(x^3 - 6x^2 - 3x + 3)$.	10
4	4-1) Write a method that takes a string as input, capitalizes the first letter of every word, and returns the result. (5 pts) 4-2) Write a method that takes a string as input, splits it into words, and returns them as an array of strings. (5 pts)	10
5	Write a program that takes a number in base 10 and prints its equivalent in base 32.	10
6	Write a program that reads 100 numbers from a file named A.txt, sorts them, and stores the result in file B.txt.	10
7	Assume the following methods already exist (i.e., you do not need to implement them – assume they exist and can be used; you may also write and use other methods of your own): BigDiv, BigMod, BigSum, BigMinus, BigMultiple, BigCompare. 7-1) Write a program that prints the result of the series $1^1 + 2^2 + 3^3 + \dots + 100^{100}$. (10 pts) 7-2) Write a program that computes and prints the 100,000th Fibonacci number. (10 pts) 7-3) Write a program that computes and prints the 100,000th prime number. (20 pts)	40

First Semester 1390

#	Question	Score
1	Write a program that takes 50 numbers, then prints the average of the numbers that are both prime and part of the Fibonacci series. Explain each part of the written program in your own words. (5 pts for not doing so)	10
2	Write a program that computes and prints the following series with precision 0.0000001. (The program must not raise an Error.) $1 + 2!/2^2 + 3!/3^3 + 4!/4^4 + \dots$ Explain each part of the written program in your own words. (5 pts for not doing so)	15
3	A method that takes a string as input, converts its lowercase letters to uppercase and its uppercase letters to lowercase, and returns the result as the method's output. Explain each part of the written program in your own words. (5 pts for not doing so)	10
4	Write a program that reads a 10×10 matrix from a file named A.txt, sorts each of the matrix's rows, and stores the result in file B.txt. Explain each part of the written program in your own words. (5 pts for not doing so)	15
5	Write a program that takes 100 numbers and stores them in an array. Then find the numbers with the most repetitions and print them along with how many times they repeat. (There may be several numbers tied for the most repetitions – the program must find and print all of them along with their repeat count.) Explain each part of the program and the variables used, in your own words. (5 pts for not doing so)	20
6	Write a program that takes a math expression, containing + - × / () and numbers (which may be multi-digit), as a string from input, then computes and prints its value. Explain each part of the program and the variables used, in your own words. (10 pts for not doing so)	30

First Semester 1392

#	Question	Score
1	Write a program that takes a number, then prints the average of the prime divisors of that number. (5 pts) Fully test the written program for the number 60. (5 pts)	10
2	Write a program that takes 10 numbers, places them in an array, then takes a number n and shifts the numbers in the array n positions forward circularly, printing the final array. (5 pts) Test the program for the numbers 3 4 65 33 12 7 44 -5 4 8 with n=3, fully step by step. (5 pts)	10
3	Write a method that takes a string as input, removes its repeated (duplicate) characters, and returns the result as the method's output. (5 pts) Fully test the written method for "tEAchEr a iS aLi". (5 pts)	10
4	Write a program that takes a number in base 10 and computes and prints its equivalent in base 32. (5 pts) Fully test the written program for the number 8345. (5 pts)	10
5	Write a program that reads a 10×10 matrix from a text file named D:\A.txt, sorts each column, and stores the resulting matrix in file D:\B.txt.	10
6	Write a method that takes a string (containing an ordinary decimal number such as "-328.098" or "101.009") as input, converts the decimal number found in the string into a floating-point number, and returns it as the method's output. (5 pts) Fully test the written method for the string "-430.008". (5 pts)	10
7	Answer only <u>one</u> of the questions in this part: 7-1) Write a program that takes a string (at most 10 characters) as input and prints all of its subsets. 7-2) Write a program that takes a string (at most 10 characters) as input and prints all of its permutations.	10
8	Assume the following methods already exist (you do not need to implement them – assume they exist and can be used; you may also write and use other methods of your own): BigDiv, BigMod, BigSum, BigMinus, BigMultiple, BigCompare. 8-1) Write a program that prints the result of the series $1^1+2^2+3^3+\dots+100^{100}$. (10 pts) 8-2) Write a program that computes and prints the (10^{100}) -th prime number. (20 pts) <i>Note: the number 10^{100} does not fit into a long or a double.</i>	30

First Semester 1393

#	Question	Score
1	Write a program that takes a number, then prints the digits of that number that are smaller than the average of its digits. (5 pts) Fully test the written program for the number 43113271. (5 pts)	10
2	Write a program that takes 10 numbers into an array. Then take a number n, and remove from the array any number whose sum of divisors equals n (removal by shifting the numbers and shrinking the array length), then print the remaining numbers. Fully and step-by-step test the program for the numbers 9 3 7 5 6 3 8 with n=6. (5 pts)	10
3	Write a method that takes a string as input and removes its repeated (duplicate) characters, then returns the result as the method's output. (5 pts) Fully test the written method for "tEAchEr a iS aLi". (5 pts)	10
4	First write a method named FullTrim that takes a string as input and removes all its extra spaces, returning the result as the method's output. (5 pts) Write a program that reads a text file named D:\A.txt (a collection of strings across several lines), removes all extra spaces from it using the method above, and stores the result in file D:\B.txt. (5 pts)	10
5	Write a program that takes a number n, then takes n numbers into an array. The numbers taken have a special property: starting from index i onward, all numbers in the array are sorted in descending order. After taking the n numbers and storing them, write a program that finds and prints i.	10
6	Write a program that takes two strings s1, s2 and prints the words common to both. What methods are needed to write this program? First name all the required methods, describe and implement each one, then write the requested program based on the methods you implemented.	10
7	Write a program that takes a sentence and removes its repeated words. (The same notes as question 6 apply here as well.)	9
8	Assume the following methods already exist (you do not need to implement them – assume they exist and can be used; you may also write and use other methods of your own): BigDiv, BigMod, BigSum, BigMinus, BigMultiple, BigCompare. Write a program that computes and prints the (10^{100}) -th prime number. (Describe each step of the program.) <i>Note: the number 10^{100} does not fit into a long or a double.</i>	20

First Semester 1394

#	Question	Score
1	Write a program that takes a number, then prints the divisors of that number that belong to the Fibonacci series. (5 pts) Fully test the written program for the number 16. (5 pts)	10
2	Write a program that takes a polynomial as a string from input, finds its highest exponent, and prints it.	10
3	Write a program that takes a math expression as a string containing ordinary decimal numbers and the signs - and +, then prints its value as a decimal number (e.g. "-438.001078-0.2021+2.3101003"). You may use the parseDouble method.	10
4	First write a method named FullTrim that takes a string as input and removes all its extra spaces, returning the result as the method's output. (5 pts) Then write a program that reads a text file named D:\A.txt (a collection of strings across several lines), removes all extra spaces from it using the method above, and stores the result in file D:\B.txt. (5 pts)	10
5	Write a method that takes a string as input and removes its consecutive repeated words, returning the result as the method's output. (Word boundaries are identified by the occurrence of a space character.)	10
6	Write a program that takes a number and prints its equivalent in base 32. (Base-32 digits: 0,1,2,3,4,5,6,7,8,9,A,B,C,...,V,10,...) (You must use a more optimized approach – using many if statements will not receive credit.)	15
7	First write the following methods: – Method setRand, which takes a 20-element array as input and fills that same array with random numbers from 1 to 100. (5 pts) – Method isSort, which takes a 20-element array as input, checks whether it is sorted, and returns the result of the check as 0 or 1. (5 pts) Then, using the methods above, write a program that fills a 20-element array with random numbers from 1 to 100, checks whether the array is sorted, and repeats this process until the generated random numbers happen to be sorted. (5 pts)	15
8	Assume all of the following methods have already been written and exist: – BigSum, BigMinus, BigMultiple, BigCompare, BigDiv, BigMod (as described in earlier sections). Write a program that computes and prints the first 1000-digit prime number.	20

Good luck and be happy – Irani